

3 (a) Product of the force on the body and displacement of the body in the direction of the force. B1

(b) (i) Work done = $Fd = (3.00 + 5.00) (0.500) = 4.00 \text{ J}$ B1

(ii) By conservation of energy,

Gain in KE = Lost in GPE(B) – Gain in GPE(A) - work done against friction M1

$$\frac{1}{2} (7.00) v^2 = 5.00 (9.81) (0.500 \sin 60^\circ) - 2.00 (9.81) (0.500 \sin 30^\circ) - 4.00$$

$$= 12.33 \text{ J} \quad \text{M1}$$

$$v = 1.88 \text{ m s}^{-1} \quad \text{A1}$$

(iii) Assuming that the static frictional force is at maximum value,

$$\text{Net force on the system} = m_B g' \sin 60^\circ - m_A g' \sin 30^\circ = 6.53 \text{ N} \quad \text{C1}$$

$$(g' = 0.20 g = 1.96 \text{ N kg}^{-1})$$

Since the net force is lesser than the maximum frictional force of 8.00 N, the net force is zero and thus, the two bodies will not move. A1

4 (a) Within the limit of proportionality, the extension/compression of a material is B1